Yihui Wang

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Ph.D. student, ECE, University of Maryland | neuromorphic and asynchorous IC design | two tapeouts, TSMC 180/65 nm

Education

University of Maryland, College Park – *Ph.D., Electrical Engineering, expected 2029, GPA 3.81/4.0* **Sep 2024 – Present**
George Washington University – *M.S. in Electrical Engineering, GPA 3.83/4.0* **Sep 2021 – Dec 2023**
Yanshan University – *B.Eng. in Photoelectron Information Science and Engineering* **Sep 2016 – Jun 2020**

Silicon Tapeouts

SpineLink: 32-Neuron Asynchronous Spiking-Neural-Network Chip **Taped out Jun 2026**
TSMC 180 nm | 1 mm^2 die, 21.5k-instance digital core | Genus, Innovus, Virtuoso, Calibre *Lead digital designer*

- **Design and implementation:** clockless tree–spine NoC of 18 bundled-data micropipeline routers, 1,152 SPI-programmed synapses, asynchronous SerDes chip-to-chip links; RTL to GDS, integration with 32 analog LIF neurons in Virtuoso, full-chip Calibre DRC/LVS.
- **Post-layout: 3.8 ns and 17 pJ per router hop**, 412 fJ per spike, 1.9 µW digital static power.

Synthesizable Asynchronous AER Encoder **Taped out May 2025**
TSMC 65 nm | TSRI shuttle | Genus, Innovus, Virtuoso, Calibre *RTL, PnR, pad frame, sign-off, bring-up*

- **Design:** 8-event bundled-data AER encoder built from standard cells only; runs through the stock synthesis and PnR flow.
- **Silicon: 33 MEvent/s, 50 ns latency, 435 fJ/event**, 8.7 µW measured; published at IEEE MWSCAS 2026.

Research Experience

Shah Lab, University of Maryland **Sep 2024 – Present**
Graduate Research Assistant – lead digital designer on the two tapeouts above *College Park, MD*

- **MURI program:** contribute to the ARO MURI on hybrid artificial–living neural networks (lead: Prof. Wolfgang Losert), replicating the learning rules of biological neurons in silicon spiking networks.
- **SpineLink NoC:** evaluated five NoC topologies against sparse spike traffic with closed-form models and synthesized routers; the resulting tree–spine NoC needs **2.9× less router area** than a 2-D mesh at 1,024 neurons and **17 vs 48 nJ per broadcast spike**.
- **Clockless methodology:** design method for standard digital flows so power scales with activity: micropipelines with automated timing closure (`set_max_delay` iteration, SDF simulation); proven on 65 nm silicon, open source.
- **FPGA spiking decoder:** 811-input, 10-neuron LIF decoder on Spartan-7 at 100 MHz, **97.9%** accuracy at **9.2 pJ per synaptic op**, 1.3k LUTs, 0 DSPs (npj manuscript).

ADAM Lab, George Washington University **Sep 2023 – Sep 2024**
Graduate Research Assistant *Washington, DC*

- **Izhikevich SNN on FPGA:** 16-bit fixed-point network with pipelined, time-multiplexed neurons (open source, 8 stars).

Projects

LIF Spiking Decoder on FPGA **2026**
Digilent Arty S7-50 (Spartan-7) | Vivado | 100 MHz *Verilog, Python*

- Designed an 811-input, 10-neuron LIF decoder: Q6.4 membrane, 4-bit weights, one event per clock, 6 Mbaud UART link.
- Achieved **97.9% accuracy** (snnTorch network reference) at **9.2 pJ per synaptic operation**; 1,323 LUTs, 0 DSPs.

Digital Design Flow (open source, 18 GitHub stars) **2025 – 2026**
Genus, Innovus, Xcelium | TSMC 180 / 65 / 28 nm, SkyWater 130 nm *Tcl, Shell*

- RTL-to-GDS Tcl flow with post-synthesis and post-route SDF simulation; used for both tapeouts and by lab members.

MIPS32 5-Stage Pipelined CPU **Nov 2023**
AMI 0.5 µm | Design Compiler, Innovus *Verilog*

- Forwarding, stall and branch-flush hazard handling; scan chains inserted for DFT; timing closed at **50 MHz**.

Publications

- **Y. Wang**, S.-Y. Peng, S. Shah, "An Ultra-Low-Power Synthesizable Asynchronous AER Encoder for Neuromorphic Edge Devices," *IEEE MWSCAS*, 2026 (accepted); arXiv:2604.05313.
- J. A. Miller, **Y. Wang**, C.-Y. Lin, S. Shah, W. Losert, "The Choice of Timescales in Hybrid Biological–Artificial Networks," *npj Unconventional Computing*, 2026 (under review).
- **Y. Wang**, S. Shah, "SpineLink: A Fully Asynchronous Tree–Spine NoC with In-Network Broadcast and Convergecast for Multi-Chip Neuromorphic Systems," *IBM IEEE CAS/EDS AI Compute Symposium*, 2026 (poster, accepted).

Technical Skills

- **Design:** RTL and asynchronous design, RTL-to-GDS, STA/SDC, DFT, DRC/LVS sign-off, FPGA, silicon test
- **EDA tools:** Cadence Genus, Innovus, Voltus, Xcelium, Virtuoso, Spectre; Synopsys Design Compiler; Calibre; Vivado
- **Languages:** Verilog/SystemVerilog, Verilog-A, Tcl, SKILL, Python (PyTorch, snnTorch), C/C++, Shell, MATLAB